

Total No. of Questions : 6]

SEAT No. :

P476

[Total No. of Pages : 2

APR - 18/BE/Insem. - 17

B.E. (Mechanical)

POWER PLANT ENGINEERING
(2012 Course) (402047) (Semester - II)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6.
- 2) Figures to the right indicate full marks.
- 3) Neat diagrams must be drawn whenever necessary.
- 4) Assume suitable data, if necessary.
- 5) Use of steam table and mollier chart and scientific calculator is allowed.

- Q1)** a) Discuss the present scenario of power generation in India. [4]
b) In a power station annual factors are as under, Load factor = 60%, capacity factor = 50% & use factor = 60%, the power plant has a maximum demand of 45,000 KW, Find :
i) Annual Energy Production.
ii) Reserve capacity above peak load.
iii) Hours per year the Plant is not in use. [6]

OR

- Q2)** a) Explain essential properties of coal required for thermal power plant. [4]
b) A thermal power plant consists of two units of 30 MW each running for 8200 Hours and one unit of 10 MW running for 2000 hours in a year. The energy produced by plant 400×10^6 KWh per annum. Determine the plant load factor and plant use factor. The Maximum demand is equal to the plant capacity. [6]

- Q3)** a) Explain with neat sketch the ash handling system. [4]
b) In a two stage steam turbine working on ideal Rankine cycle, Steam enters the first stage at 90 bar and 550°C temperature. Second stage entry temperature after reheating is 550°C. Steam is never wet during expansion in both sides. Find the cycle efficiency and specific steam consumption. Also find the steam Flow rate in the turbine in kg/sec if the turbine is directly connected to an alternator of 150 MW output. Assume 0.9 as alternate efficiency. Draw T-s diagram for the cycle. Neglect pumps work. [6]

OR

P.T.O.

Q4) a) Discuss factors to be considered in selecting the site of thermal power plant. [4]

b) In a reheat cycle, steam at 500°C Expands in a HP Turbine till it is saturated vapor. It is reheated at constant pressure to 400°C and then expands in a LP turbine to 40°C. If the maximum moisture content at turbine exhaust is limited to 15%, determine : [6]

- i) reheat pressure
- ii) boiler pressure
- iii) cycle efficiency
- iv) steam rate.

Q5) a) Write a note on hydrograph and flow duration curve. [4]

b) Write short note on : Nuclear waste disposal. [6]

OR

Q6) a) Explain classification of Hydro electric power plant. [4]

b) Explain the working of CANDU reactor with sketch. [6]

